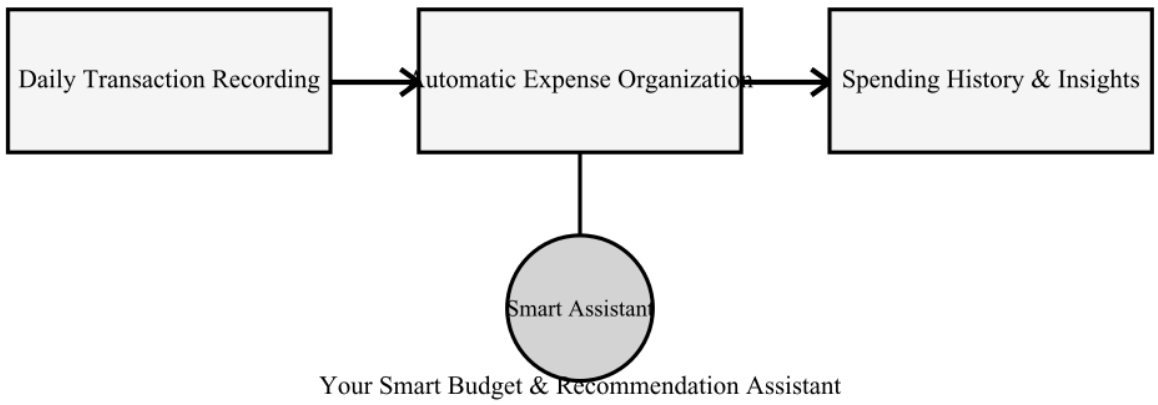
YOUR SMART BUDGET & RECOMMENDATION ASSISTANT

1.1 Difficulty in Tracking Expenses

Expanded Problem Statement



Overview

Managing personal expenses manually becomes difficult when a user makes many small transactions during the day. Food, transport, shopping, bills, subscriptions, and other purchases can quickly become difficult to remember. When records are incomplete, the user cannot accurately understand total spending or identify areas where money is being wasted.

Daily Transaction Recording

The assistant allows users to record each transaction with important information such as amount, date, description, and category. Keeping these records in one place creates a reliable history of spending and reduces the chance of forgetting small purchases.

Automatic Expense Organization

Expenses can be grouped into categories such as food, travel, shopping, education, entertainment, and utilities. Categorization makes large amounts of transaction data easier to understand and helps the user identify which categories consume the most money.

Spending History & Insights

The system maintains daily, weekly, and monthly history and can generate summaries and charts. By comparing different periods, users can identify changes in spending behavior and make better decisions about future expenses.

1.1 Difficulty in Tracking Expenses – Implementation

Sample Coding Logic

The following simple Python example demonstrates the core logic related to this problem area. It can later be integrated with a database, dashboard, notification system, and predictive model.

```
expenses = [
("Food", 450),
("Travel", 300),
("Shopping", 900),
("Bills", 1200)
]
total = sum(amount for category, amount in expenses)
print("Total expense:", total)

if total > 2500:
print("Review spending categories")
else:
print("Spending is within the tracked limit")
```

System Benefits

The feature is designed to make financial management easier, more organized, and more understandable. By converting raw expense information into useful insights, the assistant can help users take timely action.

1.1 Difficulty in Tracking Expenses – Summary

Key Points

Area	Current Challenge	Proposed Support
Tracking	Transactions may be forgotten	Record every expense in one system.
Organization	Expenses are difficult to classify	Group transactions into categories.
Analysis	Users cannot see spending trends	Provide history, summaries, and charts.

Expected Outcome

A centralized expense-tracking feature can give users a clearer view of where their money is going. It creates a foundation for budgeting, prediction, recommendations, and savings management.